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FINAL PROJECT REPORT

VN-Edu LOD

A Five-Star Linked Open Dataset of Vietnamese Higher Education Institutions

Ontology Engineering, Multi-source Integration, OWL 2 RL Reasoning,
Quality Validation and Web Publication

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[City], October 2026

Abstract

Information about Vietnamese higher education institutions is scattered across Wikipedia articles, Wikidata items and administrative documents, and it became partly obsolete when the country reorganised its provinces in July 2025. This report presents VN-Edu LOD, an open knowledge graph of **300 Vietnamese educational institutions** (271 of them higher education institutions), together with their 63 current and former provinces, 50 governing bodies and 1,695 people, published according to the *five-star Linked Open Data* scheme. We designed an OWL 2 RL ontology of 34 classes and 47 properties whose defined classes and property chains let a rule reasoner classify institutions (public/private, military/police, member institutions) and propagate locations through the 2025 province mergers; data-quality constraints are expressed separately in SHACL. A medallion (bronze/silver/gold) pipeline collects data from Wikidata and Vietnamese Wikipedia, resolves entities by Wikidata identifiers, records every conflict and fallback value, and enforces a JSON-Schema data contract before RDF generation. The resulting graph contains 39,507 triples (14,666 of them inferred), reuses standard vocabularies for 74% of asserted triples, and links 84% of its entities via `owl:sameAs` to Wikidata, DBpedia, ROR and GeoNames. Every one of its 2,241 HTTP URIs dereferences to HTML with embedded JSON-LD, Turtle and JSON-LD, and a hardened SPARQL 1.1 endpoint supports federated queries. Compared with a previous version of the project, VN-Edu LOD answers 21/21 benchmark facts correctly (vs. 14/21), is logically consistent (0 vs. 349 violations), and answers aggregate and multi-hop questions that the earlier model could not express. An empirical comparison also shows why identifier-based linking was preferred over string-similarity linking (100% vs. at most 98.3% precision, with 66–84% recall for the latter).

Keywords: Linked Open Data, knowledge graph, ontology engineering, OWL 2 RL, SHACL, data integration, `owl:sameAs`, Wikidata, higher education, Vietnam.

Contents

1	Introduction	3
2	Background and Related Work	4
2.1	Linked Data and the five-star scheme	4
2.2	Ontologies, reasoning and validation	4
2.3	Knowledge graphs extracted from Wikipedia	4
2.4	Identity links and link discovery	4
2.5	Prior course projects (baselines)	4
3	System Architecture and Methodology	4
4	Ontology Design	5
4.1	Design principles	5
4.2	Class hierarchy	5
4.3	Key axioms	6
4.4	SHACL shapes	6
4.5	URI design	6
5	Data Acquisition and Integration	7
5.1	Sources	7
5.2	Cleaning semi-structured infoboxes	7
5.3	Entity resolution and scope	8
5.4	Conflict resolution and fallbacks	8
5.5	Data contract	8

5.6	Worked example	8
6	RDF Generation and Linking	9
6.1	Vocabulary reuse and provenance	9
6.2	URI strategy: when to reuse and when to mint	9
6.3	Identifier-based linking with verification	10
7	Reasoning and Validation	10
8	Publication and Access	11
8.1	Dereferenceable URIs	11
8.2	Querying	11
8.3	Securing a public SPARQL endpoint	12
9	Evaluation	12
9.1	Dataset statistics	12
9.2	Completeness	12
9.3	Correctness against the baseline	13
9.4	Logical consistency	14
9.5	Link quality	14
9.6	Competency questions	15
9.7	Performance	15
9.8	Security	15
9.9	Five-star compliance	16
10	Discussion	16
11	Conclusion and Future Work	17
A	Namespaces	19
B	Example federated query	19
C	Reproducing the results	19

1 Introduction

Vietnam has more than 250 higher education institutions (HEIs) of very different legal types: national and regional universities, universities, academies, officer schools, and their member schools and branches. Basic facts about them – founding year, governing ministry, public or private ownership, location, leadership, admission codes – are published in many places but rarely in a form that machines can query, compare and combine. Two recent events make the problem more pressing. First, Resolution 202/2025/QH15 [21] merged the 63 provinces into 34 from 1 July 2025, so the location of many institutions in existing datasets is now obsolete. Second, community knowledge bases such as Wikidata [31] and DBpedia [18] cover Vietnamese institutions only partially and inconsistently, while the richer Vietnamese Wikipedia infoboxes are semi-structured text.

The Linked Data principles [4, 5] and the five-star open data scheme provide a well-established way to publish such information: give every entity an HTTP URI, return useful RDF when the URI is looked up, use W3C standards, and link to other datasets. This project applies them to Vietnamese higher education and, in doing so, addresses the weaknesses of an earlier version of the same project [9, 23], whose ontology produced 349 logical contradictions under reasoning and whose URIs could not be dereferenced.

Research questions. The work is organised around four questions:

- RQ1** How can an ontology for Vietnamese higher education be designed so that standard rule-based reasoning is both *correct* (no spurious contradictions) and *useful* (it derives facts users ask for)?
- RQ2** How can two heterogeneous community sources be integrated into accurate, traceable data, and how much more accurate is the result than a single-source approach?
- RQ3** How should the dataset be linked to existing Linked Data – in particular, when should existing URIs be reused and when should new URIs be minted?
- RQ4** Does the published dataset actually satisfy all five stars, and is it usable by external people and programs?

Contributions.

1. An OWL 2 RL ontology (34 classes, 27 object and 20 datatype properties) with defined classes, four property chains and disjointness axioms, aligned to schema.org, FOAF and DBpedia, plus 8 SHACL shapes (Section 4).
2. A reproducible bronze/silver/gold integration pipeline with an explicit conflict-resolution policy, provenance at the level of individual Wikipedia revisions, and a JSON-Schema data contract (Section 5).
3. A URI and linking strategy that reuses existing vocabularies and IRIs wherever possible, mints URIs only for described entities, and links them with owl:sameAs after explicit verification (Section 6).
4. A public, five-star publication: 2,241 dereferenceable URIs, a hardened SPARQL endpoint with federated queries, interactive maps, statistics and an ontology browser (Section 8).
5. A quantitative evaluation of completeness, correctness, consistency, link quality, performance and security, including a comparison with the previous version and with string-similarity linking (Section 9).

2 Background and Related Work

2.1 Linked Data and the five-star scheme

Linked Data [4, 12] rests on four principles: use URIs as names for things; use HTTP URIs so that people can look them up; return useful information in standard formats (RDF [6], SPARQL [11]) when a URI is looked up; and include links to other URIs. The five-star scheme grades open data from *available on the Web under an open licence* (★) to *machine-readable* (★★), *non-proprietary format* (★★★), *W3C standards and URIs* (★★★★) and *linked to other data* (★★★★★). The W3C Best Practices [15] and Cool URIs [27] recommend reusing vocabularies, minting stable HTTP URIs in a namespace the publisher controls, and describing datasets with VoID [2] and DCAT [1]. Empirical studies show that conformance to these practices is far from universal [13].

2.2 Ontologies, reasoning and validation

OWL 2 RL [19] is the OWL profile designed for rule-based implementation: reasoning can be performed by forward chaining in polynomial time, which makes it practical to *materialise* all consequences and serve them from an ordinary triple store. Because OWL follows the open-world assumption, cardinality restrictions do not detect missing data; SHACL [16] closes this gap by validating RDF graphs against shapes. PROV-O [17] records provenance and JSON-LD [28] embeds RDF into Web pages.

2.3 Knowledge graphs extracted from Wikipedia

DBpedia [18] and YAGO [29] extract knowledge graphs from Wikipedia infoboxes and categories; Wikidata [31] is a collaboratively edited knowledge base that now serves as a hub of identifiers. Hogan et al. [14] survey the broader field of knowledge graphs. Schema.org [8] provides widely used classes such as `CollegeOrUniversity`.

2.4 Identity links and link discovery

`owl:sameAs` asserts that two URIs denote the same individual; Halpin et al. [10] showed that it is frequently misused for mere similarity, which propagates errors under reasoning. Link discovery frameworks such as Silk [30] generate links from similarity metrics and thresholds. Zaveri et al. [32] survey Linked Data quality dimensions, several of which (completeness, consistency, interlinking, licensing, provenance) we measure in Section 9.

2.5 Prior course projects (baselines)

vio [9, 23] built a Vietnamese DBpedia-style ontology and a dataset of universities from Vietnamese Wikipedia; it is the direct predecessor of this work and serves as our main baseline. *hust-semantic-web* [24] built a smartphone dataset using a medallion architecture and Silk linking; we adopted its medallion idea and used its linking approach as a comparison point.

3 System Architecture and Methodology

Figure 1 shows the overall architecture. It follows the *medallion* pattern [7]: raw snapshots (bronze), cleaned and integrated records (silver) and RDF (gold), with a *quality gate* at every boundary so that defective data stops the pipeline instead of reaching publication.

Reproducibility. All HTTP responses are cached on disk, so re-running the pipeline yields identical output; every bronze snapshot has a `.meta.json` file recording source endpoint, retrieval time, user agent

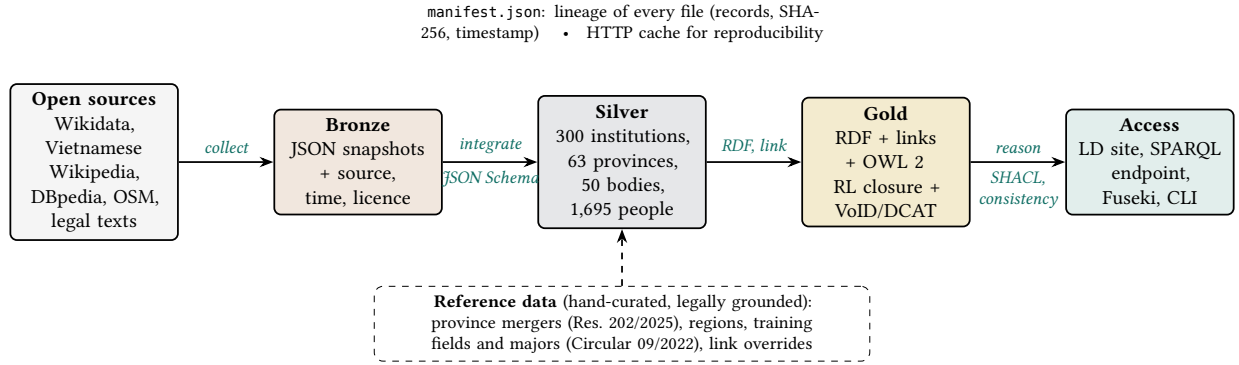


Figure 1: Architecture of the VN-Edu LOD pipeline. Each arrow is a script (step2_collect.py ... step7_publish.py); gates in italics must pass before data moves to the next layer.

and licence; and data/manifest.json records, for every file in every layer, its record or triple count and SHA-256 hash. A continuous-integration workflow runs the full test suite (100 tests) and rebuilds the published site on every commit.

Technology. The pipeline is written in Python ($\approx 5,100$ lines) using RDFLib, OWL-RL and pySHACL [26]; Apache Jena Fuseki [3] serves SPARQL locally; Oxigraph [22] compiled to WebAssembly provides SPARQL in the browser; a Flask/Waitress server provides the public endpoint.

4 Ontology Design

4.1 Design principles

The ontology (namespace vnedu:) was designed to fix the specific failures we observed when running a rule reasoner over the baseline:

1. **Stay inside OWL 2 RL** so that materialisation is sound and complete and the closure can be served by any triple store.
2. **Never redefine external vocabularies.** The baseline declared `geo:lat rdfs:domain vio:Site`; since provinces also have coordinates, the reasoner typed them as `Site`, which was disjoint with `Place` – 344 of the baseline’s 349 contradictions came from this single axiom.
3. **Align, do not equate.** Our classes are connected to schema.org, FOAF and DBpedia only through `rdfs:subClassOf/rdfs:subPropertyOf` (26 alignment axioms), never `owl:equivalentClass`, so external semantics cannot leak in.
4. **Property chains range over common super-classes**, so that chaining never forces an individual into an incompatible class.
5. **Constraints belong in SHACL**, because OWL’s open-world semantics cannot detect missing values.
6. **Defined classes for classification, disjointness and functional properties for error detection.**

4.2 Class hierarchy

Figure 2 shows the class hierarchy. Institution types follow the Law on Higher Education [20] and are determined from the institution’s official name (*Đại học*, *Trường Đại học*, *Học viện*, *Trường Sĩ quan*) rather

than from Wikidata’s P31, which mixes universities, colleges and faculties. The four legal types are pairwise disjoint (`owl:AllDisjointClasses`). Six *defined classes* are populated by the reasoner only.



Figure 2: Class hierarchy of the vnedu: ontology (34 classes). Classes in orange italics are *defined* classes whose members are inferred by the reasoner. Organization, Person and AdministrativeUnit are mutually disjoint, as are the four legal HEI types; Branch and VocationalCollege are disjoint with HigherEducationInstitution. Alignments ($\sqsubseteq$): Organization to `foaf:Organization` and `schema:Organization`; HigherEducationInstitution to `schema:CollegeOrUniversity`; University and UniversitySchool to `dbo:University`; Person to `foaf:Person` and `schema:Person`; Major and FieldOfStudy to `skos:Concept`.

4.3 Key axioms

Table 1 lists the axioms that drive reasoning, each with a real example from the data. All are covered by unit tests that check both the positive inference and the detection of erroneous data.

4.4 SHACL shapes

Eight node shapes with 23 property constraints and 2 SPARQL-based constraints check what OWL cannot: language-tagged labels in Vietnamese and English, `xsd:gYear` founding years between 1800 and the current year, admission codes matching `^[A-Z]{3}$`, coordinates inside Vietnam’s bounding box, exactly one region per province, member institutions in the same province as their parent (informational), and a warning when a private institution lists a ministry as its governing body.

4.5 URI design

All URIs live under the namespace controlled by the project, `https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/`, with the pattern `resource/{type}/{slug}`, where the slug is the accent-free official Vietnamese name (e.g. `resource/university/dai-hoc-bach-khoa-ha-noi`). Slugs are stable (they do not encode database keys) and human-readable; homonyms of defunct institutions receive their period of activity, e.g. “Đại học Cần Thơ (1966–1975)”. The ontology itself is published at `.../ontology` with hash URIs for terms.

Table 1: Main OWL 2 RL axioms and what they derive in the dataset.

Axiom (DL notation)	Effect on real data
$locatedIn \circ mergedInto \sqsubseteq locatedIn$	Thu Dau Mot University is recorded in Bình Dương (former province) $\Rightarrow$ located in Ho Chi Minh City.
$locatedIn \circ partOf \sqsubseteq locatedIn, partOf$ transitive	$\Rightarrow$ located in Nam Bộ (region) and Vietnam.
$governedBy \circ subordinateTo \sqsubseteq governedBy$	Naval Academy is governed by the Navy $\Rightarrow$ governed by the Ministry of National Defence.
$MilitaryInstitution \equiv$ $EduOrg \sqcap \exists governedBy. \{MoND\}$	$\Rightarrow$ 21 military institutions (7 police institutions analogously).
$\exists ownedBy. Company \sqsubseteq$ $\exists ownership. \{Private\}$	FPT University is owned by FPT Education $\Rightarrow$ private.
$PublicInstitution \equiv$ $EduOrg \sqcap \exists ownership. \{Public\}$	$\Rightarrow$ 221 public / 35 private institutions.
$rector, director, councilChair \sqsubseteq$ $hasLeader \equiv leads^-$	$\Rightarrow$ 214 <i>InstitutionLeaders</i> .
$offersProgram \circ ofMajor \sqsubseteq trainsMajor$	Majors taught by an institution derived from its programmes.
8 inverse pairs (<i>memberOf/hasMember</i> , <i>alumnusOf/hasAlumnus</i> , ...)	Only one direction needs to be asserted.
<i>ownership, foundingYear</i> functional; $AllDifferent(Public, Private)$; legal types $AllDisjointClasses$;	Detect contradictory data: an institution both public and private, two founding years, both an academy and a university, or a person used as a governing body.
$Organization \sqcap Person \sqsubseteq \perp$	

5 Data Acquisition and Integration

5.1 Sources

Table 2 summarises the bronze layer. Wikidata is queried with one SPARQL query per property using VALUES blocks (avoiding Cartesian products and time-outs); Vietnamese Wikipedia is crawled recursively through 164 categories (e.g. *Đại học Việt Nam*, *Học viện Việt Nam*) and through every article linked from Wikidata, so that institutions missing from categories are not lost.

Table 2: Bronze-layer snapshots.

Snapshot	Source	Records	Licence
Institution facts (labels, types, inception, website, coordinates, ROR, leaders, ...)	Wikidata SPARQL	363	CC0
Infoboxes, lead sections, <i>History</i> sections	Vietnamese Wikipedia API	279	CC BY-SA 4.0
Image/logo metadata (URL, licence, author)	Wikipedia/ Commons API	280	per file
Organisations linked from infoboxes (title $\rightarrow$ QID)	Wikipedia API	80	CC BY-SA 4.0
Notable alumni (gender, occupation, birth date)	Wikidata SPARQL	1,531	CC0
Provinces (population, area, GeoNames)	Wikidata SPARQL	63	CC0
Founding years (cross-check)	DBpedia SPARQL	72	CC BY-SA 3.0
Province mergers, regions, training fields	Res. 202/2025, Circ. 09/2022	curated	public law

5.2 Cleaning semi-structured infoboxes

Infobox values are wikitext, not data. The parser (based on `mwparserfromhell`) handles line-break templates whose removal would glue tokens together (`BKHN<br>HUST` $\rightarrow$ “`BKHN | HUST`”), list templates, date templates (`{{start date|1956|3|6}}` $\rightarrow$ `1956-03-06`), degree-minute-second coordinates, references,

invisible soft hyphens and image captions. Infobox templates with English parameter names (principal, parent, affiliation) are normalised to their Vietnamese equivalents; before this normalisation one institution (VinUni) silently lost its rector, owner, council chair and partners. Person names are separated from academic titles and ranks (e.g. “GS.TS.NGND” → vnedu:honorific).

5.3 Entity resolution and scope

Wikidata QIDs are the primary key. Missing sitelinks are repaired by title search (accepting only articles not attached to another item); three groups of duplicate Wikidata items (e.g. Hanoi Law University has three QIDs) are merged into one entity that keeps owl:sameAs to all of them. **68 candidate entities were excluded**, each with a recorded reason: faculties and departments, high schools, list and history articles, and English-only items that could not be cross-checked.

5.4 Conflict resolution and fallbacks

Table 3 states the resolution policy per field. All 42 cases where sources disagree are written to conflicts.csv; every value obtained from a lower-priority fallback (176 values, mostly coordinates, governing bodies and ownership) is written to filled.csv with its source, so a reviewer can audit exactly which facts were not taken from a structured field.

Table 3: Source priority per field (first available wins; disagreements are logged).

Field	Priority
Province	infobox city/province → Wikidata P131 → headquarters address → Wikidata P159/P276 → lead sentence near “trụ sở/đặt tại” → reverse geocoding of coordinates (OSM) → place name in the official name
Founding year	earliest of Wikidata, infobox and DBpedia (institutions date themselves from their predecessor); then “thành lập năm ...” in the lead, taking the year nearest to the keyword
Ownership	infobox type → Wikidata class (public/private university) → strict pattern in the lead; plus the OWL rule for company-owned institutions
Leaders	infobox (usually more current) → Wikidata statements without end date (P582)
Coordinates	Wikidata → infobox → geocoded address, accepted <i>only</i> if inside the chosen province
Relations	a “member of / under” target is typed by what it resolves to: HEI → vnedu:memberOf/vnedu:branchOf; ministry or committee → vnedu:governedBy; company → vnedu:ownedBy

5.5 Data contract

Before RDF generation every silver record is validated against a JSON Schema (file schemas/silver.schema.json) covering identifiers, enumerations, value ranges, URL and email formats, image URLs, and cross-field rules (e.g. a former province must have a successor). Any violation stops the pipeline.

5.6 Worked example

Figure 3 follows VinUni through the layers: the infobox (bronze), the cleaned record (silver) and the RDF (gold). The founding year (2018, approval of the project, from Wikidata) and the founding date (17 December 2019, licensing, from the infobox) are both kept, and the difference is logged as a conflict rather than silently resolved.

Bronze – infobox wikitext (English parameter names)

```
| other_name = VinUniversity      | established = {{Start date and age|2019|12|17}}
| parent = [[Tập đoàn Vingroup]] | affiliation = [[Đại học Cornell]]{{.}}[[Đại học
→ Pennsylvania]]{{.}}...
| chairman = Lê Mai Lan          | principal = Tan Yap-Peng      | image = Trường Đại học VinUni
→ logo.png
```

Gold – RDF (excerpt), reusing standard vocabularies

```
university:truong-dai-hoc-vinuni a vnedu:UniversitySchool ;
  rdfs:label "Trường Đại học VinUni"@vi , "VinUni"@en ; skos:altLabel "VinUniversity" ;
  schema:foundingDate "2019-12-17"^^xsd:date ; vnedu:foundingYear "2018"^^xsd:gYear ;
  vnedu:ownership vnedu:PrivateOwnership ; vnedu:ownedBy organization:tap-doan-vingroup ;
  vnedu:rector person:tan-yap-peng ; vnedu:councilChair person:le-mai-lan ;
  dbo:affiliation wd:Q49115 , wd:Q49117 ; # partners = existing Wikidata IRIs
  schema:logo <https://upload.wikimedia.org/wikipedia/vi/4/4b/...VinUni_logo.png> ;
  owl:sameAs wd:Q86012411 , dbr:VinUniversity , <https://ror.org/052dmdr17> ;
  prov:wasDerivedFrom wd:Q86012411 , <https://vi.wikipedia.org/w/index.php?oldid=75478634> .
```

Figure 3: One institution through the pipeline. After reasoning the entity additionally becomes a `vnedu:PrivateInstitution`, a `schema:CollegeOrUniversity` and is located in `Bắc Bộ` and `Vietnam`.

6 RDF Generation and Linking

6.1 Vocabulary reuse and provenance

RDF is generated from silver records by `step3_transform.py`. Standard vocabularies are used wherever an appropriate term exists – RDFS, SKOS, FOAF, schema.org, DBpedia ontology, W3C Geo, PROV-O – so that **74% of asserted triples** (15,083 of 20,446) use externally defined predicates. Every institution carries `prov:wasDerivedFrom` links to its Wikidata item *and to the exact Wikipedia revision* (oldid) that was parsed, so any value can be traced to the text it came from. Textual descriptions (`dbo:abstract`, `vnedu:history`) are attributed to that revision under CC BY-SA.

6.2 URI strategy: when to reuse and when to mint

A natural question is why the project mints its own URIs instead of using Wikidata’s. Our answer follows the W3C best practices [15]:

- **Reuse whenever we only *refer* to something.** Occupations and genders of alumni, partner universities and image files are referenced directly by their existing IRIs (222 Wikidata IRIs in 2,523 triples), with only a label added for display. We do not mint `vnedu:occupation/politician`.
- **Mint only for entities we *describe*.** The 2,241 entities we publish facts about need URIs whose lookup returns *our* description (Linked Data principle 3): admission codes, legal type, post-2025 province, inferred classification. A Wikidata URI cannot be made to return that. Keeping our assertions under our own URIs also keeps their provenance separate, covers entities Wikidata does not have (training majors, province-merger relations), and lets the project guarantee dereferenceability of its namespace.
- **Assert identity explicitly.** 84% of minted entities (1,874/2,241) carry `owl:sameAs` to existing URIs; where identity is not exact (a training *major* vs. an academic *discipline*) we use `skos:closeMatch` instead, following [10].

6.3 Identifier-based linking with verification

Links are derived from shared identifiers rather than name similarity (Table 4). Wikidata links come from the data’s own provenance; ROR and GeoNames links from Wikidata properties P6782 and P1566; DBpedia links from DBpedia’s own `owl:sameAs` to Wikidata. Two verification steps guard against wrong identity links: governing bodies and regions without a QID are searched by Vietnamese label *and* must satisfy `P17 = Vietnam` (this rejected, e.g., a Korean ministry with the same name); and DBpedia resources that DBpedia maps to several Wikidata items are rejected (14 cases, e.g. `dbp:Pleiku` wrongly mapped to Gia Lai province). Links are not fed to the reasoner, because `owl:sameAs` semantics would copy every triple onto the external URIs.

Table 4: Outgoing links (gold layer), described as `void:Linksets` in the dataset description.

Target	Predicate	Links	Derivation
Wikidata	<code>owl:sameAs</code>	1,880	provenance; label search + country check
Wikipedia (vi, en)	<code>foaf:isPrimaryTopicOf</code>	1,188	sitelinks
DBpedia	<code>owl:sameAs</code>	260	DBpedia <code>owl:sameAs</code> → QID, ambiguity filter
ROR	<code>owl:sameAs</code>	195	Wikidata P6782
GeoNames	<code>owl:sameAs</code>	63	Wikidata P1566
Wikidata, DBpedia	<code>skos:closeMatch</code>	88	majors; exact label + manual overrides

7 Reasoning and Validation

The asserted data, ontology and VoID description (21,071 triples) are materialised with OWL-RL [26] in about 30 seconds, producing a closure of 59,880 triples. From the closure we keep only *instance-level* triples whose subject belongs to the dataset (14,666 new triples), which avoids publishing tautologies about the vocabularies. A set of SPARQL checks then looks for `owl:Nothing` members, violations of disjointness and `AllDifferent`, and functional-property clashes: **the dataset is consistent (0 violations)**. SHACL validation of asserted plus inferred data **conforms** and reports 37 warnings and 1 informational result (Table 5); the warnings are genuine gaps in the open sources, deliberately not filled by guesswork.

Table 5: SHACL validation report (severity $\geq$ Warning does not block publication).

Message	Count	Severity
Missing founding year	22	Warning
Missing province of headquarters	12	Warning
Private institution lists a ministry as governing body	3	Warning
Member institution located in a different province from its university	1	Info

Test suite. 100 automated tests run in continuous integration: 14 reasoning tests (each axiom of Table 1, five tests that inject erroneous data and expect a contradiction, and one test asserting that the real dataset is consistent), 29 extraction tests using real Wikipedia sentences, 3 data-contract tests, 11 infobox/media tests, and 43 web-server tests (content negotiation, SPARQL protocol, security).

8 Publication and Access

8.1 Dereferenceable URIs

Every one of the 2,241 URIs resolves on GitHub Pages to (i) an HTML page for people, with an embedded JSON-LD description, a Wikipedia-style infobox, introduction and history text, a mini-map and markers on every inferred value; (ii) .ttl and (iii) .jsonld representations announced by `<link rel="alternate">` (Figure 4). Because GitHub Pages is static, content negotiation is provided by a second deployment of the same pages behind a Flask/Waitress server (`https://vnedu-lod.onrender.com`), which returns HTML, Turtle, JSON-LD, N-Triples or RDF/XML according to the Accept header. The dataset is described with VoID and DCAT (licence, sources, linksets, distributions, SPARQL endpoint).

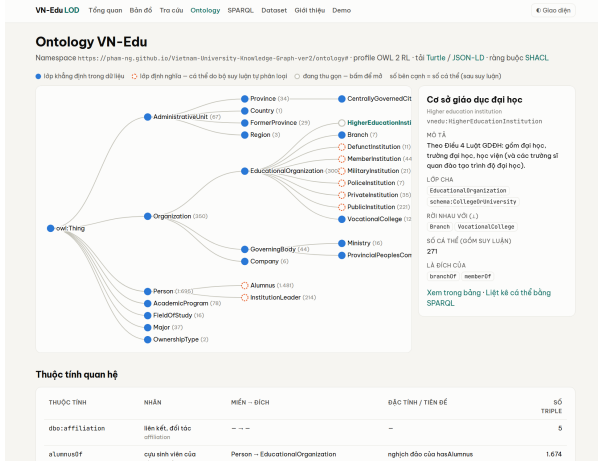
The screenshot shows the website for VinUni (Trường Đại học VinUni). The top navigation bar includes links for VN-Edu LOD, Tổng quan, Bản đồ, Tra cứu, Ontology, SPARQL, Dataset, Giới thiệu, and Demo. A 'Giáo diện' button is in the top right. The main content area has a header with the university's name and a URL: `https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/resource/university/truong-dai-hoc-vinuni`. Below the header, there is a section titled 'Giới thiệu chung' (General Introduction) and another titled 'Lịch sử' (History). The 'Giới thiệu chung' section describes the university's founding and its affiliation with the VinGroup. The 'Lịch sử' section details the university's development from 2017 to 2020. On the right side, there is an infobox titled 'Trường Đại học VinUni' with a logo and a table of key information.

Trường Đại học VinUni	
TÊN KHÁC	VinUniversity
TÊN TIẾNG ANH	VinUni
LOẠI HÌNH	Trường đại học · Tư thực
THÀNH LẬP	17 tháng 12 năm 2019 (mốc sớm nhất giữa các nguồn: 2018)
TỔ CHỨC MẸ	Tập đoàn Vingroup
HIỆU TRƯỞNG	Tan Yap - Peng
CHỦ TỊCH HỘI ĐỒNG TRƯỞNG	Lê Mai Lan
ĐỐI TÁC	Đại học Cornell Wikidata Đại học Pennsylvania Wikidata Học viện Công nghệ Georgia Wikidata Đại học Quản lý Singapore Wikidata Đại học Công nghệ Nanyang Wikidata
ĐỊA CHỈ	Vinhomes Ocean Park, Đa Tốn, Gia Lâm

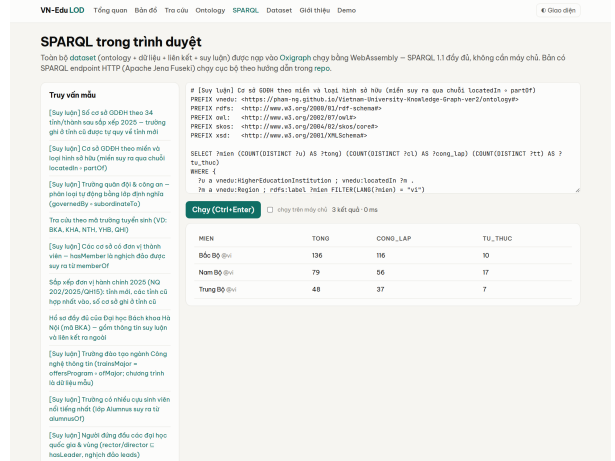
Figure 4: The URI of VinUni looked up in a browser: introduction and history text (with revision-level attribution), and an infobox generated from the RDF (logo, legal type, founding date, owner, rector, council chair, partners linked to Wikidata).

8.2 Querying

Three query channels are offered: (1) a public SPARQL 1.1 Protocol endpoint (`https://vnedu-lod.onrender.com/sparql`) that supports federated queries [25] to Wikidata and DBpedia; (2) SPARQL in the browser, where Oxigraph compiled to WebAssembly loads the full dataset client-side (Figure 5b); and (3) Apache Jena Fuseki plus a command-line client for local use. Fifteen documented competency queries ship with the dataset (Table 9).



(a) Interactive ontology browser with instance counts.



(b) SPARQL in the browser: institutions per region and ownership, where the region is inferred.

Figure 5: Ontology and query interfaces of the published site.

8.3 Securing a public SPARQL endpoint

An open endpoint executes arbitrary user queries, so we audited the server and reproduced two real vulnerabilities in the first version: a **server-side request forgery** (a query with `SERVICE <http://127.0.0.1:...>` made the server issue requests to internal addresses) and a **SPARQL injection** through the resource path. The hardened server only allows `SERVICE` calls to an allow-list (Wikidata, DBpedia), rejects variable endpoints and `FROM` clauses, validates resource paths against the dataset’s URI pattern before building queries, limits query length (20,000 characters) and execution time (30 s), returns generic error messages, and sets security headers. A further practical issue was that RDFLib hard-codes a User-Agent that Wikimedia rate-limits; the server replaces it with a descriptive one so that federated queries work. All of these behaviours are covered by tests.

9 Evaluation

9.1 Dataset statistics

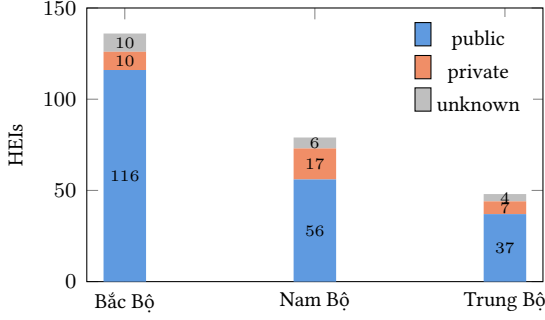
Table 6 and Figure 6 summarise the dataset.

Table 6: Dataset size.

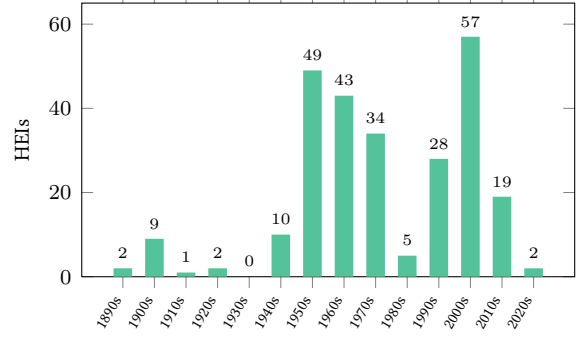
Entities	Count	RDF	Triples
Educational institutions	300	Asserted data	20,446
of which HEIs	271	Outgoing links	3,674
Provinces (current / former)	34 / 29	Inferred (instance level)	14,666
Governing bodies and companies	50	VoID/DCAT	96
People (alumni / leaders)	1,481 / 214	Ontology	629
Majors / fields / programmes	37 / 16 / 78	Total served	39,507
Dereferenceable URIs	2,241	Entities with <code>owl:sameAs</code>	84%

9.2 Completeness

Figure 7 reports the share of the 300 institutions that have each property. Identity and classification properties are nearly complete; scale figures (students, staff) and admission codes are sparse because the



(a) HEIs by region (inferred via *locatedIn* $\circ$ *partOf*) and ownership.



(b) HEIs by decade of (earliest recorded) foundation.

Figure 6: Two views of the data, both produced by SPARQL over the inferred graph. Hà Nội (103) and Ho Chi Minh City (58) host 60% of HEIs.

open sources rarely provide them – a limitation we report rather than fill with unverifiable values.

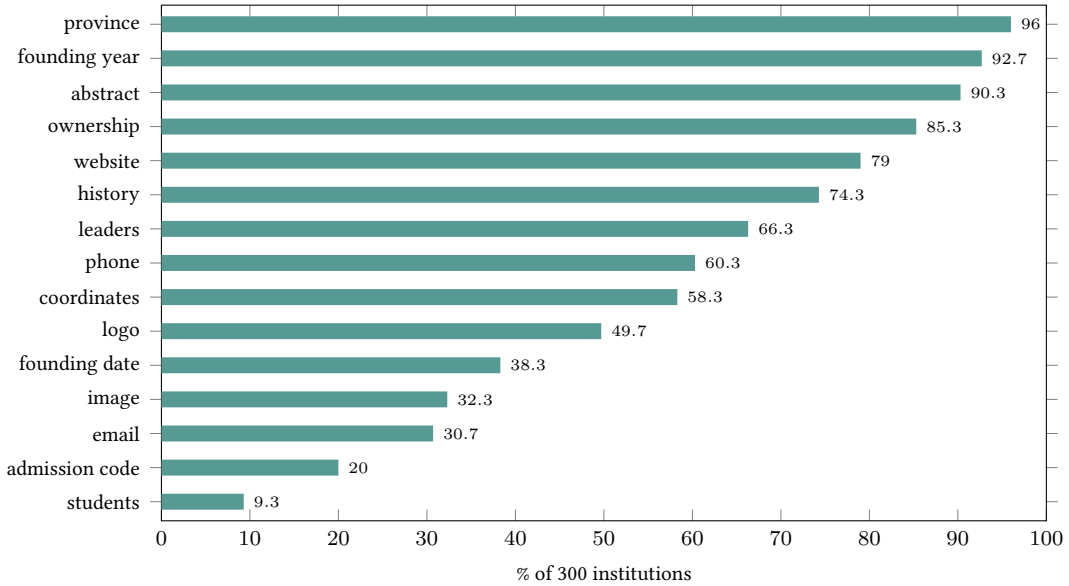


Figure 7: Property completeness over the 300 institutions (asserted data).

9.3 Correctness against the baseline

We built a benchmark of 21 facts with verified answers (12 founding years, 5 headquarters provinces after the 2025 reform, 4 governing bodies, including ones requiring inference) and ran both datasets through the same OWL 2 RL reasoner. VN-Edu LOD answers **21/21** correctly, the baseline **14/21** (Table 7). The baseline errors are typical of single-source extraction from free text: a year mentioned in the article but unrelated to the founding (Hanoi Law University “1727”; Naval Academy “2022”), missing entities, and an obsolete or wrong province.

Beyond these facts, aggregate questions reveal modelling errors in the baseline: it lists 62 members of Vietnam National University, Hanoi (true: 12, because “member of” and “governed by the Ministry” were not distinguished) and 40 military schools (21 after correctly typing governing bodies), and it cannot express public/private ownership, regions, dissolved institutions or admission codes at all.

Table 7: Benchmark of 21 verified facts (both datasets reasoned with the same engine).

Category	Baseline	VN-Edu LOD	Examples of baseline errors
Founding year (12)	6	12	Hanoi Law Univ. 1727 (true 1979); Naval Academy 2022 (1955); An Giang Univ. 1976 (1999); Hanoi Nat. Univ. of Education 1950 (1951); 2 institutions missing
Province after 2025 (5)	4	5	Thu Dau Mot Univ. placed in Bình Phước (true: Bình Dương → Ho Chi Minh City)
Governing body (4)	4	4	–
Total (21)	14	21	

9.4 Logical consistency

Under OWL 2 RL materialisation the baseline produces **349 contradictions** (344 from the redefined domain of `geo:lat`, 5 institutions with two founding years); VN-Edu LOD produces **none**. The baseline also equated its classes with external ones (`owl:equivalentClass` to `dbo:University` and `schema:CollegeOrUniversity`) and declared a sub-property of `owl:sameAs`, which is not allowed in OWL DL.

9.5 Link quality

Identifier-based vs. string-similarity linking. To justify the linking strategy we simulated a Silk-style [30] matcher – Jaccard similarity over lower-cased tokens of English names, best candidate above a threshold θ – on 86 HEIs whose correct DBpedia resource is known from identifiers (Table 8). At the threshold used by the hust-semantic-web baseline ($\theta = 0.2$) one link in six is wrong; raising θ buys precision at a steep cost in recall. The root cause is structural: Vietnamese university names share most tokens. *Hanoi University of Science and Technology* and *University of Science and Technology of Hanoi* are two different institutions with *identical* token sets (Jaccard = 1.0); no threshold separates them, whereas shared identifiers (QID, ROR) do.

Table 8: String-similarity linking vs. identifier-based linking on 86 HEIs (ground truth from identifiers).

Method	θ	Links	Wrong	Precision	Recall	F1
Token Jaccard (Silk-style)	0.2	85	13	84.7%	83.7%	0.842
	0.4	80	8	90.0%	83.7%	0.867
	0.6	76	6	92.1%	81.4%	0.864
	0.8	68	4	94.1%	74.4%	0.831
	1.0	58	1	98.3%	66.3%	0.792
Shared identifiers (this work)	–	86	0	100%	100%	1.000

Link validity. We dereferenced a random sample of 30 links per target (150 links). All 150 returned HTTP 200; Wikidata, DBpedia and GeoNames returned RDF when RDF was requested; 30/30 ROR records were the correct Vietnamese organisation; and 29/30 Wikidata labels matched (the remaining item is the correct entity under a former name).

External usability. Acting as an external client, we parsed the live Turtle and embedded JSON-LD of sample resources (identical, 192 triples for HUST), followed all 162 internal URIs mentioned on that page (162/162 dereferenceable), and parsed 40 randomly sampled resources out of 2,241 (40/40 valid RDF).

9.6 Competency questions

Table 9 lists the documented queries. Nine of them depend on inferred triples; two are federated and retrieve facts the dataset does not store (province area from Wikidata to compute HEI density; logos and mottos from DBpedia), which is the practical benefit of the owl:sameAs links.

Table 9: Competency questions shipped with the dataset (queries/*.rq).

#	Question	Inference	Example answer
1	HEIs per post-2025 province (old-province records mapped to new)	✓	Hà Nội 103, HCMC 58
2	HEIs per region $\times$ ownership	✓	Bắc Bộ 136 (116 public)
3	Military and police institutions	✓	21 + 7
4	Lookup by admission code (BKA, KHA, ...)		BKA $\rightarrow$ HUST, public, Hà Nội
5	Universities and their member institutions	✓	VNU Hanoi: 12
6	Old provinces merged into each new province		Res. 202/2025
7	Full profile of HUST including inferred facts and links	✓	192 triples
8	Institutions training IT, by region	✓	(sample programme data)
9	Institutions with most notable alumni	✓	ranked list
10	Leaders of national and regional universities	✓	5 universities
11	Link statistics per target dataset		Table 4
12	Federated: HEI density per 1,000 km ² using Wikidata areas		Hà Nội 30.7
13	Federated: logo, motto, students from DBpedia		HUST motto
14	Training-field catalogue with skos:closeMatch		16 fields
15	Data quality: share of HEIs with each property		Figure 7

9.7 Performance

Table 10 reports measured latencies. In the browser the whole dataset (39,507 triples) loads in 2.3 s and the competency queries run in 2–34 ms. On the public server (free tier, single shared CPU, rdflib in memory, 121 MB resident memory, 3.5 s start-up) typical queries take 1–3.5 s and URI dereferencing about 0.2 s. Federated queries are dominated by the remote endpoints.

Table 10: Measured latency (median of 5 runs unless noted).

Channel	Operation	Time
Browser (Oxigraph/WASM)	load 39,507 triples	2.3 s
	competency queries 1–10	2–34 ms
Public server (rdflib)	Q01 per-province aggregation	1.7 s
	Q02 region $\times$ ownership	1.1 s
	Q09 alumni ranking	3.5 s
	dereference URI as Turtle / HTML	0.18 / 0.22 s
	Q12 federated (Wikidata) / Q13 (DBpedia), single run	5.9 / 8.7 s
Local server (Waitress)	20 concurrent dereferences	0.83 s total

9.8 Security

All attacks reproduced against the first server version were re-run against the deployed endpoint and are rejected: SERVICE to a cloud metadata address and to localhost (HTTP 400), SPARQL injection through the resource path (404), over-long queries (413) and path traversal (400). Normal federated queries to Wikidata and DBpedia still succeed.

9.9 Five-star compliance

Table 11: Five-star assessment of the published dataset.

Star	Requirement	Evidence
★	On the Web, open licence	Public at https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/ ; CC BY-SA 4.0 (required by Wikipedia share-alike) declared in DCAT, licence file and every bronze snapshot
★★	Machine-readable	JSON, CSV, RDF
★★★	Non-proprietary formats	Turtle, N-Triples, JSON-LD, CSV
★★★★	W3C standards, URIs	RDF, OWL 2 RL, SHACL, SPARQL 1.1, PROV-O, VoID, DCAT, JSON-LD; 2,241 HTTP URIs that dereference (verified externally)
★★★★★	Linked to other data	3,674 links to Wikidata, DBpedia, ROR, GeoNames, Wikipedia, verified (150/150 sample); federated queries exploit them

10 Discussion

Lessons learned. (1) *Most reasoning failures were design failures, not data failures:* a single domain declaration on an external property caused 98% of the baseline’s contradictions. Keeping the ontology in OWL 2 RL, aligning instead of equating, and moving completeness checks to SHACL removed all of them. (2) *Two independent sources are worth more than one large source:* most baseline errors were values that one source states and another contradicts; logging conflicts makes these visible. (3) *For identity, identifiers beat similarity:* with name-similarity matching the best achievable F1 on our data was 0.867, while identifiers gave exact links. (4) *Publishing is engineering:* the first public endpoint had an SSRF and an injection vulnerability, and federated queries failed because of a third-party User-Agent policy; neither was visible until the system was tested as an external client would use it.

Limitations.

- **Coverage is bounded by the open sources.** There is no machine-readable official register of Vietnamese HEIs, so completeness cannot be proven; private institutions (35 identified) are probably under-represented, and 24 HEIs have unknown ownership.
- **Sparse quantitative attributes:** students (9%), admission codes (20%), staff counts.
- **Programmes are sample data:** 37 majors and 78 programmes were curated by hand for 16 institutions; queries about “who trains major X” are illustrative, not statistics.
- **Limited temporal modelling:** leader terms, dated student counts and renaming events are not modelled as time-indexed facts.
- **Hosting:** primary URIs are on static hosting; content negotiation and the SPARQL endpoint run on a free-tier server that sleeps when idle and keeps the graph in memory, which suits a dataset of this size but not heavy load. Timed-out queries cannot be forcibly stopped by rdflib.

Threats to validity. The correctness benchmark is small (21 facts) and biased towards well-known institutions, so it demonstrates the kind of errors avoided rather than estimating an error rate for the whole dataset; a random sample of about 50 institutions checked against official websites would give such an estimate. Wikipedia and Wikidata are community-edited and may be outdated (e.g. leaders);

revision-level provenance makes every value auditable but does not make it correct. Server latencies were measured on a shared free-tier instance and vary with load.

11 Conclusion and Future Work

VN-Edu LOD shows that a small team can publish a genuinely five-star dataset by combining careful ontology design, multi-source integration with explicit provenance and quality gates, identifier-based linking, and an engineering approach to publication. The resulting knowledge graph of 300 institutions and 39,507 triples is logically consistent, answers all benchmark facts correctly, supports aggregate, multi-hop and federated questions, and is usable by external clients through dereferenceable URIs and a public SPARQL endpoint.

Future work includes: integrating the Ministry of Education and Training’s official lists and admission data to measure and improve coverage; modelling time-indexed facts (leadership terms, enrolment by year, renaming and merger events) with an n-ary or RDF-star pattern; replacing the sample programme data with programme catalogues; registering a persistent namespace (e.g. `w3id.org/vnedu`) with server-side content negotiation; and contributing verified corrections (e.g. missing sitelinks and ambiguous DBpedia mappings found in this work) back to Wikidata.

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A Namespaces

Prefix	IRI
vnedu:	https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/ontology#
university:, province:,...	https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/resource/...
schema:	https://schema.org/
dbo:	http://dbpedia.org/ontology/
dbr:	http://dbpedia.org/resource/
wd:	http://www.wikidata.org/entity/
prov:	http://www.w3.org/ns/prov#

B Example federated query

```
# HEI density per province: our data + area fetched from Wikidata through owl:sameAs
PREFIX vnedu: <https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
SELECT ?tinh (COUNT(DISTINCT ?u) AS ?so_co_so) ?dien_tich_km2 WHERE {
  ?u a vnedu:HigherEducationInstitution ; vnedu:locatedIn ?p .
  ?p a vnedu:Province ; rdfs:label ?tinh ; owl:sameAs ?wd .
  FILTER(LANG(?tinh) = "vi" && STRSTARTS(STR(?wd), "http://www.wikidata.org/entity/"))
  OPTIONAL { SERVICE <https://query.wikidata.org/sparql> { ?wd wdt:P2046 ?dien_tich_km2 } }
} GROUP BY ?tinh ?dien_tich_km2 ORDER BY DESC(?so_co_so)
```

C Reproducing the results

```
pip install -r requirements.txt
powershell -File run_all.ps1          # collect -> integrate -> RDF -> link -> reason -> report ->
  -> publish -> tests
python -m pytest tests -q              # 100 tests
python app/server.py --prod            # local server with the same interface; Fuseki:
  -> fuseki/run_fuseki.ps1
curl -H "Accept: text/turtle"
  -> https://vnedu-lod.onrender.com/resource/university/truong-dai-hoc-vinuni
```